

Information Gathering Tools

The tool family

Tool	Purpose
nmap	The Swiss-army scanner (host discovery, ports, services, OS)
netdiscover / arp-scan	Layer-2 host discovery on a live subnet
theHarvester	Collect emails, subdomains, hosts from search engines
recon-ng	Modular OSINT framework (modules for many sources)
Maltego	Graph-link OSINT (entities + transforms)
dig / nslookup / host	DNS interrogation
whois	Domain registration records
Masscan / RustScan	Very fast port sweeps (scan first, nmap second)

Rightsize your scanning

```
# Host discovery
netdiscover -r 10.0.0.0/24
arp-scan --localnet

# FAST port sweep with Masscan
sudo masscan -p1-65535 --rate=1000 10.0.0.20 -oL open.txt

# Focused nmap on discovered ports
nmap -sS -sV -sC 10.0.0.20 -oA target

# DNS
dig any example.com
nslookup -type=TXT example.com
whois example.com
```

```
# Emails / subdomains
```

```
theHarvester -d example.com -b google,linkedin,crtsh
```

Reading results

Open ports → services → versions → attack candidates

arp-scan reveals live hosts the firewall might not (L2 bypasses L3 filtering)

Recon-ng keeps your OSINT modular and repeatable

Speed vs thoroughness

Scan tiers: (1) fast sweep for open ports, (2) version detection on open ports, (3) script/exploit checks on interesting ones. Skipping to full-version on everything wastes hours.

OPSEC your scanning

Scanning is *loud*. It is also fully authorized in your lab – the point is to learn what it sounds like to defenders.

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